ONTOLOGY OF TRANSITION—PART II

The Unidentifiable Clock: Reconstruction Limits and Gauge Freedom of External Time under Lossy Delivery

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Abstract

This is Part II of a series. Part I [1] argued that a system's accessible clock coordinate is a functional of a physically transmitted, generally lossy record of a reference process, not a reading of the source itself — a conceptual organization of established results. Part II turns that qualitative thesis into quantitative theorems about the same model, using Part I's record, register, and decoder without modification. First, for an erasure channel with known tick count, the minimax mean-squared error of reconstructing the source coordinate from the delivered record is exactly n p σ^2 — the number of source ticks times the loss probability times the variance of the calibration increments — achieved by the conditional-mean decoder; in particular, the source coordinate is perfectly recoverable through an arbitrarily lossy channel iff calibration is deterministic. Second, when the tick count itself is unknown, an additional floor of μ^2 n p/(32 π q) is proved for deterministic calibration by a self-contained two-point argument. Third, and centrally: for the aggregating channel, the pair (loss rate, calibration law) is never identifiable from the record law alone for any positive loss rate — it is determined only up to a one-parameter semigroup orbit of geometric compounding, which contains at most one geometrically indecomposable representative; when the physical calibration law is indecomposable, it is that representative. Operationally: an observer behind the channel cannot distinguish a fast source behind a lossy channel from a slow source behind a clean one. The rate of external time is gauge, and the gauge freedom is generated by geometric compounding. Fourth, the reader itself is bounded: for a register in a nonequilibrium steady state, the relative variance of the decoded coordinate is at least 2/reg by the thermodynamic uncertainty relation — the accessible clock thus carries two distinct irreducible error floors, expressed in different metrics and attributed to different physical mechanisms — one information-theoretic (paid in lost calibration fluctuation), one thermodynamic (paid in register dissipation). Fifth, objecthood itself has a critical point: for the canonical alternating source read through the aggregating channel, robust token recognition in the sense of Part I's recognition scheme survives exactly up to a critical loss rate pc = $(\delta{+}\varepsilon)/(1{-}\delta{-}\varepsilon)$, set by the channel's parity bias, above which the faster type dissolves while the frozen type persists at any loss. Provenance side-information collapses the gauge orbit and restores identifiability, exactly as the parent framework's message-order protocol suggests. All results are verified numerically.

Keywords: *external clocks; lossy delivery; reconstruction limits; gauge freedom; thermodynamic uncertainty; object recognition*

1. Setting

We work inside the model of Part I [1, §3.2, App. A], keeping its notation; the record-as-message viewpoint descends from the event-ordering tradition of [2]. A related formal treatment of mediated interaction, execution traces, and observational metadata is developed in [13]. The setting: a clock process C emits ticks c_1, c_2, ... with calibrated increments k > 0 (eq. (26) of [1]); a delivery channel produces a record of attempts with provenance sets Aj (eq. (A1)); the system's decoded coordinate accumulates decoded increments (eq. (27)). Throughout, the calibration increments k are i.i.d. from a law F on (0, ∞) with mean $\mu < \infty$ and variance $\sigma^2 < \infty$. Two channel archetypes from [1, App. A] are formalized:

(E) Erasure channel Each tick is independently delivered with probability q = 1 − p and lost with probability p (its index belongs to no Aj). Delivered ticks are decoded exactly and in source order. The record after n source ticks is the ordered list of delivered increments; the observer sees the list and its length N, nothing else.

(A) Aggregating channel Ticks are processed in order; each is independently buffered with probability p or flushed with probability q, a flush delivering one message that carries the sum of all buffered increments since the previous delivery, plus its own, decoded exactly (eq. (A3), aggregate case). Message values are then i.i.d. compound-geometric: the m-th block has size distributed Geom(q) on {1, 2, . . . }, and value distributed G = m1 q pm−1 F*m.

The target of reconstruction is the source coordinate tC(n) = kn k (eq. (26) of [1]). The observer's information is the σ-algebra generated by the record — the register history FnR of [1, eq. (A2)].

2. The exact reconstruction floor

Theorem II.A (price of a lost tick). Under channel (E) with known n and known (p, F), for every estimator t measurable with respect to the record,

$$Et - tCn2np2$$

with equality for the conditional-mean decoder t* = SN + (n − N) μ, where SN is the sum of delivered increments and N their number.

Proof. Write tC = SN + M, where M is the sum of the lost increments. The delivery mask is independent of the increment values, and the increments are i.i.d.; hence, conditionally on the record (which determines N and the delivered values), M is a sum of n − N i.i.d. draws from F, independent of the delivered values. Therefore E[M | record] = (n − N) μ and Var(tC | record) = Var(M | record) = (n − N) σ^2 almost surely. For any record-measurable t, E[(t − tC)^2] ≥ E[Var(tC | record)] = σ^2 E[n − N] = n p σ^2, and the conditional mean attains it. □

Corollary II.A1 (dichotomy of recoverability). tC is exactly recoverable from the record for some (equivalently, every) p ∈ (0, 1) iff σ^2 = 0, i.e. iff calibration is deterministic. Losing ticks is not the same as losing time: with a degenerate calibration the decoder t reconstructs tC without error at any loss rate, because each lost tick carries a known duration. What a lossy channel irrecoverably destroys is exactly the fluctuation of calibration, at a rate of σ^2 per lost tick.

Remark. Var(tC | record) is precisely the variance of the filtering distribution Π of [1, eq. (A2)]; Theorem II.A computes it in closed form for channel (E) and shows the parent framework's "state of knowledge" is not a metaphor but the exact Bayes object.

Numerical check Appendix N.1: empirical MSE matches the floor to 0.3%; the degenerate calibration gives identically zero error.

3. Unknown tick count

Lemma II.B1 (local bound). maxi P(Bin(n, q) = i) ≤ $\sqrt{}$(π / (8 n q p)), where p = 1 − q.

Proof. By Fourier inversion, every point mass is bounded by (1/2π)∫−π^π |ϕ(t)| dt, where ϕ is the characteristic function. For the binomial, |ϕ(t)|^2 = 1 − 2qp(1 − cos t) ≤ exp(−2qp(1 − cos t)), hence |ϕ(t)| ≤ exp(−n q p (1 − cos t)). Using 1 − cos t ≥ 2t^2/π^2 on [−π, π] and extending the Gaussian integral to the line, (1/2π)∫ exp(−2nqp t^2/π^2) dt = (1/2π) · 3/2/$\sqrt{}$(2nqp) = $\sqrt{}$(π/(8nqp)). □

Theorem II.B (counting floor; local minimax). For channel (E), suppose σ^2 = 0, the tick count n is unknown to the observer, and n p ≥ 2πq. Then every record-measurable estimator obeys the bound below. The explicit constant 1/(32π) is not claimed to be optimal.

$$supn'n, n + kEt - tCn'22np/32q$$

Proof. With $\sigma^2 = 0$ the record law depends on n only through N $\sim$ Bin(n, q) (delivered values are i.i.d. F under both hypotheses and carry no information about n). Step 1 (shift TV): for the unimodal binomial, TV(X, X+1) $= \frac{1}{2}\ \Sigma_i\ |p_i - p_{i-1}|$ = maxi p_i by telescoping over the mode, so TV(X, X+j) $\leq$ j $\cdot$ maxi p_i; by the coupling Bin(n+k, q) =d Bin(n, q) + Bin(k, q) and convexity, TV(Bin(n,q), Bin(n+k,q)) $\leq$ E[Bin(k,q)] $\cdot$ maxi $p_i \leq$ kq $\cdot \sqrt{}(\pi/(8nqp))$ = k $\cdot \sqrt{}(\pi q/(8np))$ by Lemma II.B1. Step 2 (two points): choose k = $\lfloor\sqrt{}(2np/(\pi q))\rfloor$; the hypothesis np $\geq 2\pi$q gives $\sqrt{}(2np/(\pi q)) \geq 2$, hence k $\geq \frac{1}{2}\sqrt{}(2np/(\pi q))$ and TV $\leq \frac{1}{2}$. The targets differ by kμ; Le Cam's two-point bound gives max risk $\geq$ (kμ/2)^2 $\cdot$ (1 $-$ TV)/2 $\geq$ k$^2\mu^2$/16 $\geq \mu^2$np/(32πq). ☐

Remark (combining the floors). For unknown n, Theorem II.A's floor persists at every σ^2: an estimator ignorant of n is admissible in each known-n problem, so the supremum over n of the risk is at least n p σ^2 pointwise in n. By the Bayes decomposition E(t $-$ tC)^2 = E(t $-$ E[tC | record])^2 + E[Var(tC | record)], the calibration floor lives in the second term for every σ^2, while the counting floor of Theorem II.B bounds the first term as proved here only at $\sigma^2 = 0$ (the two-point argument uses the determinism of the target; for $\sigma^2 > 0$ the target's own fluctuation $\sigma\sqrt{}$n is of the same order as the separation). Hence risk $\geq$ n p σ^2 always, and additionally risk $\geq \mu^2$ n p / (32πq) at $\sigma^2 = 0$; a sharp counting floor for $\sigma^2 > 0$, and whether the floors add, are left open.

Numerical check Appendix N.2: Lemma II.B1 and the shift bound verified in four regimes; the prescribed k* yields TV = 0.30 $\leq \frac{1}{2}$.

Combined with Theorem II.A: the proven floors for unknown n are n p σ^2 at every σ^2, and additionally c μ^2 n p/q when calibration is deterministic; their joint sharp form is open. In both regimes the channel destroys information at a rate linear in n p.

4. The compounding gauge: non-identifiability of the clock channel

The deeper question is not reconstruction of a trajectory but identifiability of the mechanism: does the record law determine (p, F)?

Lemma II.C0 (composition of geometric blocking). For block-size generating functions ga(z) = a z / (1 $-$ (1$-$a) z) on {1, 2, ...}: gq″ $\circ$ g = gq″ . Consequently, a Geom(q″)-sum of independent Geom(θ)-blocks is a Geom(q″ θ)-block.

Proof. Direct computation: gq″(g(z)) = q″ θ z / (1 $-$ (1 $-$ q″ θ) z). ☐

Theorem II.C (gauge orbit of external time). Under channel (A), let G(p, F) denote the law of message values. Then for every q″ $\geq$ q (i.e. every less lossy channel p″ $\leq$ p),

$$Gp, F = Gp'', F'', \qquad F'' = Geomq/q''F$$

Hence the record law determines (p, F) only up to the totally ordered orbit

$$Op, F = p'', Geomq/q''F : 0p''p$$

whose endpoint p″ = 0 declares the channel lossless and attributes all blocking to the source. Conversely, two representations (p_1, F_1), (p_2, F_2) of the same record law with $q_1 < q_2$ satisfy F_2 = Geom(q_1/q_2)$\circ F_1$. Therefore:

(i) (p, F) is never identifiable from the record law when p > 0;

(ii) the orbit contains at most one representative whose calibration law is geometrically indecomposable (not itself a nontrivial geometric compound), and if F is geometrically indecomposable then (p, F) is that unique maximal-loss representative. The orbit O(p, F) exhausts the equivalence class of the record law iff F is geometrically indecomposable; for decomposable F the class extends to representations with $p'' > p$.

Proof. In Laplace transforms, $G = q\ F\ /\ (1 - p\ F)$. For $q'' \geq q$ set $F'' = \theta\ F\ /\ (1 - (1-\theta)\ F)$ with $\theta = q/q'' \in (0, 1]$; by Lemma II.C0 this is the transform of the Geom(θ)-compound of F, a bona fide law on $(0, \infty)$, and substitution gives $q''\ F''\ /\ (1 - p''\ F'') = G$. For the converse, solving $q_1F_1/(1-p_1F_1) = q_2F_2/(1-p_2F_2)$ for F_2 yields $F_2 = \theta\ F_1\ /\ (1 - (1-\theta)\ F_1)$ with $\theta = q_1/q_2$, i.e. $F_2 = \mathrm{Geom}(q_1/q_2)\circ F_1$, which is a nontrivial compound whenever $q_1 < q_2$. Claim (ii) follows: if F_2 were also geometrically indecomposable, the displayed identity forces $q_1 = q_2$ and $F_1 = F_2$. □

Corollary II.C1 (rate of external time is channel-relative). Since the mean message value is μ/q, an observer behind channel (A) cannot distinguish "fast source, lossy channel" from "slow source, clean channel": every point of the orbit is an observationally complete description. In the vocabulary of [1], the accessible external clock fixes duration only relative to a chosen decomposition into source and channel; the decomposition itself is gauge. This sharpens the parent paper's claim that external time is relative to the triple (S, C, K): the (C, K) split inside the triple is not empirical but conventional, up to the geometric-compounding semigroup action.

Terminology Here gauge means observational redundancy among parameterizations of the same record law. It is an analogy to gauge freedom as redundancy of description, not a claim of a local symmetry of the underlying dynamics.

Corollary II.C2 (what breaks the gauge). Any side-channel that reveals provenance — stored tick numbers or causal labels in the sense of [1, App. A.2] — collapses the orbit to a point: with singleton provenance sets observable, N and the delivered indices are known and (p, F) is identifiable (F from delivered values, p from index gaps). Identifiability of the clock is thus purchased exactly by the metadata whose transmission the parent framework prices separately (§11 of [1]).

Numerical check Appendix N.3: two orbit points are statistically indistinguishable (KS $p = 0.50$); the transform identities hold to machine precision.

5. The thermodynamic floor of the reader

The thermodynamic cost of clock precision is studied theoretically and experimentally in [5, 6, 10, 11]. The theorem below transfers the current-precision bound to the reader's side of the channel, where the parent framework's separation of source, channel, and register makes the attribution exact.

Theorem II.D (register-precision bound). Let the register be a finite Markov jump process in a nonequilibrium steady state satisfying local detailed balance, whose net registration count N(t)—forward minus reverse transitions across designated registration edges—is a time-antisymmetric integrated empirical current with mean $\langle N\rangle > 0$, and let the register's total entropy production over the observation window be reg (in units of kB). Assume the decoded increments wj are i.i.d. with mean $w > 0$, independent of the registration process. Then the decoded coordinate t = jN wj satisfies

$$Var t/\langle t\rangle 2 Var N/\langle N\rangle 2 2/reg$$

Proof. By the law of total variance with w independent of N: $\mathrm{Var}(t) = \langle N\rangle\ \mathrm{Var}(w) + w^2\ \mathrm{Var}(N) \geq w^2\ \mathrm{Var}(N)$, while $\langle t\rangle = w\ \langle N\rangle$; dividing gives the first inequality. The second is the finite-time thermodynamic uncertainty relation for integrated currents in a NESS under local detailed balance [4, 7]. □

Corollary II.D1 (two irreducible floors). The accessible clock coordinate carries two distinct irreducible error budgets, expressed in different metrics and paid to different parties. The channel floor of Theorem II.A is information-theoretic: n p σ^2 of variance destroyed per window, payable in lost calibration fluctuation, independent of any thermodynamics. The register floor of Theorem II.D is thermodynamic: relative variance at least 2/reg, payable in the reader's own dissipation, independent of source quality and channel fidelity. The two budgets constrain different observables through different mechanisms; they become additive only within an explicitly specified joint error model, and neither can be traded against the other. In the vocabulary of [1]: partial observability and the physical cost of observation (§11) bound the same coordinate from two irreducible directions.

Scope of the current observable A one-sided count of registration events is activity-like and is not covered by the thermodynamic uncertainty relation used here; such an observable requires an appropriate kinetic uncertainty relation. The literal success count may be identified with N only in an effectively unidirectional registration model or when reverse events are negligible on the stated scale.

Remark (scope). The independence hypothesis is substantive. It holds for erasure-type registers — channel (E) with exact decoding is the model case, since there each decoded weight is a fresh draw from F regardless of how many registrations the window contains. For the aggregating channel (A) it fails by construction: wj is a block sum, and over a fixed window larger blocks mean fewer registrations, so decoded weights and the counting current are negatively correlated. Theorem II.D therefore does not apply to channel (A) as stated; the aggregating case is open, and any extension must replace the total-variance step by one that tracks the weight–count covariance.

Remark (attribution). The bound involves reg only. A perfect source read through a perfect channel by a dissipationless register in steady state would still be TUR-bound; conversely, an arbitrarily dissipative register cannot repair the channel floor of Theorem II.A. The parent framework's insistence on separating source, channel, and register (eqs. (19)–(24), (71)–(73) of [1]) is exactly what makes this attribution well-posed.

Numerical check Appendix N.4: a unicyclic register realizes the full chain of inequalities within 4% of TUR saturation.

6. The dissolution of objects

We now give the recognition scheme R* of [1, §12] its first sharp threshold. Take the canonical alternating source: the emitted symbol flips at every source step (the alternating regime of [1, App. B] in its pure form). (Throughout this section q = 1 − p is the flush probability of channel (A).) Read it through channel (A), observing the current symbol at each delivery. Formally, each delivered message is an observation cut in the sense of [1, Table B1], and the observed symbol is the decoded register reading of [1, App. A] immediately after the aggregated registration — for a symbolic source, the source symbol at the block's final tick.

$$f := 1/(1 + p)$$

Lemma II.E1 (parity lemma). Successive delivered observations flip iff the intervening block size is odd; the flip indicators are i.i.d. Bernoulli(1/(1+p)).

Proof. Block sizes are i.i.d. Geom(q) on $\{1, 2, \dots\}$, and an alternating source changes symbol over a block iff the block is odd. P(m odd) = j0 q p2j = $q/(1 - p^2) = 1/(1+p)$; independence is inherited from the blocks. □

As $p \to 0$ the observed sequence alternates perfectly; as $p \to 1$ the flip probability tends to $\frac{1}{2}$ and the observed sequence tends to a fair coin: the channel converts structure into noise, with the parity bias $1/(1+p)$ as the exact carrier.

Theorem II.E (critical loss for objecthood). Let the feature be the flip fraction F over a window of h consecutive flip indicators (and therefore h+1 delivered observations), the alternating type region Balt = $[1-\delta, 1]$ with $\delta \in (0, \frac{1}{2})$, and recognition thresholds drawn from $E \subseteq [0, \varepsilon]$ with $s := \delta + \varepsilon < \frac{1}{2}$. Set

$$pc = s/1 - s, \qquad f = 1/1 + p, \qquad d* = max0, 1 - -f$$

(a) (Dissolution.) If $p > pc$, then $d^* > \varepsilon$ and each window is recognized at some threshold $t \leq \varepsilon$ with probability at most $\exp(-2h(d^* - \varepsilon)^2)$. Consequently, for every fixed token length W, the probability of robust recognition over any threshold interval contained in $[0, \varepsilon]$ vanishes exponentially as $h \to \infty$.

(b) (Persistence.) If $p < pc$, then $d^* < \varepsilon$, and for every $0 < \gamma < \varepsilon - d^*$ a W-window token is recognized simultaneously at all thresholds $t \in [d^* + \gamma, \varepsilon]$ with probability at least $1 - W \exp(-2h\gamma^2)$. Robust segmentation persists with a robustness range of length at least $\varepsilon - d^* - \gamma$.

(c) (Asymmetry.) The frozen type has flip fraction identically 0 and is recognized at every threshold for every $p < 1$.

Proof. By Lemma II.E1, $hF \sim \mathrm{Bin}(h, f)$. A window is recognized at threshold t iff d(F, Balt) $\leq t$, i.e. iff $F \geq 1 - \delta - t$. (a) For every $t \leq \varepsilon$ this requires $F \geq 1 - s$, and $p > pc \iff f < 1 - s$; Hoeffding gives $P(F \geq 1 - s) \leq \exp(-2h((1-s) - f)^2)$ with $(1-s) - f = d^* - \varepsilon$. (b) $p < pc \iff f > 1 - s \iff d^* < \varepsilon$. Simultaneous recognition on $[d^* + \gamma, \varepsilon]$ is implied by recognition at the leftmost threshold, i.e. by $F \geq 1 - \delta - d^* - \gamma$; whether $d^* > 0$ (where $1 - \delta - d^* = f$) or $d^* = 0$ (where $1 - \delta \leq f$), the condition is implied by $F \geq f - \gamma$, and $P(F < f - \gamma) \leq \exp(-2h\gamma^2)$; a union bound over W windows completes it, and threshold-monotonicity of the recognized set makes the segmentation constant on the interval; no frozen window is spuriously admitted anywhere on it, since the frozen feature is 0 and its distance to Balt equals $1 - \delta > \frac{1}{2} > \varepsilon$. (c) A constant source delivers a constant symbol under any block parity. □

Corollary II.E1 (fast objects dissolve first). The critical tolerance budget satisfies $\delta + \varepsilon = p/(1+p)$: what must be spent to keep recognizing the fast object equals exactly the channel's parity bias. Since the frozen type survives every loss rate, the segmentation of the pair degrades only through its faster member: in a lossy clock, objecthood is lost in order of intrinsic speed.

Remark (beyond period two). For a period-r source the parity mechanism generalizes: the observed phase performs a random walk on Zr with step law m mod r, $m \sim \mathrm{Geom}(q)$, and recognition statistics are governed by that walk's spectral gap; the critical constant becomes a function of the gap rather than of the parity bias alone. The sharp constant is left to future work.

Numerical check Appendix N.5: the transition is bracketed on both sides of $pc = 1/3$, with critical slowing at the boundary and monotone convergence in h.

7. Further results (stated, not proved here)

Conjecture F (Markov-modulated sources). If the source is Markov-modulated (increments emitted from hidden states), identifiability of the hidden clock structure from the record is expected to reduce to a Kruskal-rank condition [3] on the emission and transition factors, by an argument analogous to standard tensor-decomposition identifiability proofs. The gauge of Theorem II.C then acts on the identifiable equivalence class rather than on a point.

8. Positioning

What this framework does not claim (i) That time, work, and information are one physical field — they are distinct typed functionals of a shared causal carrier. (ii) That continuous time necessarily emerges from noise — it enters by interpolation, or as a limit of refining clocks. (iii) That Maxwell's demon is hereby dispelled — its thermodynamics is settled by the Szilard–Landauer–Bennett line, on which Part I builds. (iv) That Newtonian time is refuted — only that it is not required as an ontological primitive of this construction. (v) That the question of thinghood is closed — the definition offered is operational and recognition-scheme-relative.

Part I develops the formal architecture — transition, causal order, internal and external clocks, and the delivery channel — while distinguishing that contribution from the established thermodynamic and information-theoretic results it uses. Part II derives quantitative consequences of that architecture. Theorem II.A identifies the exact-constant reconstruction floor and the recoverability dichotomy of Corollary II.A1; the constant is p σ^2 per source tick, turning Part I's "the record is partial" into a rate. Theorem II.C establishes the principal identifiability result: compound-geometric non-identifiability belongs to the established theory of geometric random sums and geometric infinite divisibility [8, 9]. The contribution here is its clock-channel interpretation: an explicit observational-equivalence orbit for the source/channel decomposition, its indecomposable representative, and provenance-based gauge fixing. To the best of our knowledge, that combined operational interpretation has not previously been stated for a delivered physical clock record. The resulting observational-equivalence orbit generated by geometric compounding shows that the source rate is not identifiable independently of the channel. Theorem II.E extends the analysis from time reconstruction to object recognition: the recognition scheme that Part I introduced descriptively acquires a sharp phase boundary, with the critical constant expressed entirely in the scheme's own tolerances.

The sharp critical loss rate established in Theorem II.E is structurally related to the critical observation scale introduced in [12], where model-based regulation becomes effective only above a scale-dependent threshold. The control parameters and observables are different: [12] varies smoothing scale and explained variance, whereas the present result varies channel loss and the robustness of object-token recognition.

Appendix N. Numerical verification

All simulations use NumPy's PCG64 generator; the seeds, sample sizes, and target quantities are reported below to support independent reproduction.

N.1. Theorem II.A

Seed 7. n = 200, p = 0.3, F = Exp(1) ($\mu = \sigma^2 = 1$), $2 \cdot 10^5$ trials: empirical MSE of the conditional-mean decoder 60.18 against the exact floor n p $\sigma^2 = 60.00$ (ratio 1.003). Degenerate calibration $\Delta \equiv 1$: maximal absolute error 0.0 over $2 \cdot 10^4$ trials.

N.2. Lemma II.B1 and Theorem II.B

Seed 11. Lemma II.B1 verified at $(n, q) \in \{(1000, 0.7), (200, 0.3), (5000, 0.9), (50, 0.5)\}$: e.g. bound 0.0432 against true maximum 0.0275 at (1000, 0.7). Shift bound $\mathrm{TV}(\mathrm{Bin}(n,q), \mathrm{Bin}(n+k,q)) \leq kq\sqrt{}(\pi/(8nqp))$ verified at $k \in \{5, 15, 30\}$; the prescribed $k^* = 16$, in the $(n, q) = (1000, 0.7)$ regime, yields $\mathrm{TV} = 0.2996 \leq \frac{1}{2}$.

N.3. Theorem II.C

Seed 7. $q_1 = 0.4$ versus $q_2 = 0.8$ with $F'' = \mathrm{Geom}(0.5)\circ\mathrm{Exp}(1)$, $4 \cdot 10^5$ samples per side: matching means 2.501 / 2.500 (theory $1/q_1 = 2.5$) and variances 6.245 / 6.253; two-sample KS statistic 0.0037, p-value 0.50 on 10^5-subsamples. Laplace identities at $s \in \{0.3, 1.0, 2.7\}$: discrepancy $\leq 1.1 \cdot 10^{-16}$.

N.4. Theorem II.D

Seed 11. Unicyclic three-state register, forward/backward rates $2 : 1$, with N defined as the net winding current, $T = 300$, 4000 Gillespie trials: $\mathrm{Var}(N)/\langle N\rangle^2 = 0.00998$ against $2/\Sigma = 0.00962$ (within 4% of TUR saturation); with $w \sim \mathrm{Exp}(1)$ the full chain realizes as $0.01356 \geq 0.00998 \geq 0.00962$.

N.5. Lemma II.E1 and Theorem II.E

Seeds 5, 9. Flip frequencies match $1/(1+p)$ to four digits at $p \in \{0.2, 0.5, 0.8\}$, lag-1 correlations below 10^{-3}. With $\delta = 0.10$, $\varepsilon = 0.15$ ($pc = 1/3$), margins $\gamma = (\varepsilon - d^*)/2$, $W = 20$ windows: robust-recognition probability 1.000 at $p = 0.15$ ($h = 800$), 0.952 at $p = 0.25$ ($h = 2000$), 0.657 at $p = 0.30$ ($h = 8000$; critical slowing), identically 0 at $p \in \{0.35, 0.45, 0.60\}$, where the admissible interval has negative length. Convergence in h at $p = 0.25$: 0.029, 0.195, 0.619, 0.949, 0.999 at $h = 200, 500, 1000, 2000, 4000$.

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